

Response to Reviewers — Round 2

Manuscript: Spatial Cluster Randomized Trials — Sampling Design with Spillover Effects & Spatial Dependence

Journal: BMC Medical Research Methodology

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Revised Manuscript File: Revisions_V2.tex

Dear Editors (Michael Grayling & Vanshali Mota) and Reviewer,

We thank the Editor and the Reviewer for their continued thorough and constructive engagement with our manuscript. We have carefully considered each of the six major concerns raised and have revised the manuscript accordingly. Below we provide a detailed point-by-point response to each comment. All changes are implemented in the revised file Revisions_V2.tex. Page and line numbers refer to the revised manuscript.

The revised manuscript preserves the core scientific contribution---comparing Simple Random Sampling and Block Stratified Sampling for spatial CRT design---while substantially improving the precision and clarity of the model specification, notation, and inferential framing in response to the Reviewer's concerns.

Point-by-Point Responses

Comment (ID)	Reviewer Comment (Summarized)	Location & Evidence
C1	The parameter beta is described as the 'total treatment effect' in Section 2.3.1. In the SAR model, beta represents only the direct treatment effect; the indirect spillover effect is captured by psi, and the true total effect involves both components. This mislabeling is conceptually important.	Section 2.3.1, ~line 283 in Revisions_V2.tex

Comment (ID)	Reviewer Comment (Summarized)	Location & Evidence
Response:	We thank the reviewer for this important clarification. We have revised Section 2.3.1 to state explicitly that β represents the direct treatment effect---the change in outcome attributable to treatment receipt, independent of any indirect spillover contribution. The sentence that previously described β as a 'total treatment effect' has been rewritten: we now clarify that in TrtSpill scenarios the structural model ensures $\hat{\beta}$ retains its interpretation as the direct effect, while the spillover contribution to treated units is captured separately by $\hat{\psi}$, and that the total treatment effect for a treated cluster receiving spillover is therefore $\hat{\beta} + \hat{\psi}$. This distinction is aligned with the decomposition already discussed in relation to R2.M2 from the first revision round.	
C2	Spillover is described as $\psi(d_i)$, a function of distance from the 'nearest intervention source,' but (a) 'intervention source' is never defined (cluster boundary? treated unit? individual?), (b) the model later treats spillover as a binary Rook-neighbor indicator creating inconsistency, and (c) ψ is used as a scalar in model (6). Individual-level locations are simulated but may be discarded if spillover operates only at the cluster level.	Section 2.4 SAR model, ~line 529; clarification of cluster-level indexing added ~line 527

Comment (ID)	Reviewer Comment (Summarized)	Location & Evidence
Response:	We agree that the $\psi(d_i)$ notation was misleading and have corrected it throughout. The simulation uses exclusively binary spillover based on Rook contiguity: $z_i = 1$ if cluster i shares a Rook edge with at least one treated cluster (a cluster assigned to receive the intervention), and 0 otherwise. The distance-function notation $\psi(d_i)$ has been replaced with the scalar ψ; 'intervention source' is now explicitly defined as 'treated cluster'; and z_i is defined as a binary Rook-neighbor indicator. Individual-level observations within each cluster share the cluster-level treatment status x_i and spillover status z_i---these are constant for all subjects k within cluster i. We retain Appendix E for the distance-based neighbor extension to clarify it is a generalisation, not the simulation's implementation.	
C3	Multiple notation issues: w_{ij} is not defined at Equation (1); y is indexed by i and k but x and ϵ are not; Equations (4)–(6) drop the cluster index without explanation; the W matrix definition is ambiguous regarding whether it is cluster-level or individual-level.	Section 2.4: after Eq (1) ~line 520; model description ~lines 525–530

Comment (ID)	Reviewer Comment (Summarized)	Location & Evidence
Response:	We have addressed all seven sub-items systematically. (3a) w_{ij} undefined in Equation (1). An inline definition has been added immediately after Equation (1): w_{ij} are elements of the row-normalized spatial weights matrix W (equal to $1/ N(i)$ if clusters i and j share a Rook edge, and 0 otherwise; full construction in Appendix A). (3b) Inconsistent indexing: y indexed by both i and k in Eq (1) but x and ϵ are not; x and z have i, k indexes in Eq (3). We have harmonized the indexing throughout. In Equation (1), ϵ_i has been corrected to ϵ_{ik}, consistent with y_{ik}. In Equation (3), x_{ik} and z_{ik} have been corrected to x_i and z_i, reflecting that treatment and spillover status are cluster-level quantities shared by all subjects k within cluster i. The error term ϵ_i in Eq (3) has also been corrected to ϵ_{ik}. The notation now consistently distinguishes cluster-level covariates (subscript i only) from individual-level outcomes and errors (subscript ik). (3c) Cluster index dropped in Equations (4)–(6). The ik indexing is now carried through all equations, including the spatial dependence term (Eq 4), the estimation equation (Eq 5), the data-generating expression, and the Appendix B likelihood expressions. This makes the multi-level structure explicit and consistent throughout. Within each cluster, because x_i and z_i are cluster-level constants, individual responses y_{ik} differ only through the error term ϵ_{ik}. (3d) W matrix level ambiguity. W is now described explicitly as a cluster-level $n \times n$ adjacency matrix, with subscripts i and j in w_{ij} referring to clusters rather than individuals. The description clarifies that because x_i and z_i are cluster-level constants, all individuals within a cluster share the same conditional expected outcome; within-cluster variation arises solely through ϵ_{ik}. (3e) ψ used as both a function $\psi(d_i)$ and a scalar. The function notation $\psi(d_i)$ has been replaced with the scalar ψ consistently throughout, reflecting the binary Rook-neighbor spillover mechanism actually implemented in the simulation. (3f) Redundant $N(i)$ indexing in Equation (4)	

Comment (ID)	Reviewer Comment (Summarized)	Location & Evidence
C4	The paper describes the data-generating process (DGP) but never explicitly states which model is fitted in the simulation. If the spillover indicator z_i is omitted from the estimation model this must be stated; if included, the near-collinearity between x_i and z_i needs to be addressed.	Section 3.1 after Equation DataGen, new paragraph ~line 607 in Revisions_V2.tex
Response:	We have added a new paragraph immediately after the DGP equation (Equation DataGen) that explicitly states the estimation model. The estimation model (new Eq. EstModel) has the same structural form as the DGP: $y_i = \rho \sum_j w_{ij} y_j + \alpha + \beta x_i + \psi z_i + \epsilon_i$, estimated via <code>lagsarlm()</code> MLE. We clarify that the DGP and estimation model share the same structure; any bias in $\hat{\beta}$ therefore does not arise from deliberate misspecification. Rather, it reflects (1) multicollinearity between x_i and z_i, which varies between SRS and BSS allocations, and (2) the interaction of background autocorrelation ρ with the spatial configuration of treatment assignments. The existing Limitations section already acknowledges these collinearity challenges.	
C5	BSS has only 2--6 valid allocations (nearly deterministic). The response from Revision 1 states 'BSS is a design-stage tool evaluated through exhaustive enumeration, not a randomization distribution for permutation inference,' but this does not fully address the frequentist inference problem. If treatment allocation is nearly deterministic, the permutation-based randomization test for BSS would have almost no power. What practical value does BSS have if it cannot be used for inference?	Section 3.1 inference paragraph expanded ~line 663; new sentence in Limitations ~line 1224

Comment (ID)	Reviewer Comment (Summarized)	Location & Evidence
Response:	We appreciate this important challenge and have expanded our treatment of it. The key distinction is between design selection (using BSS vs. SRS to choose an allocation) and inference (how to construct confidence intervals and p-values). We have expanded the design-stage inference paragraph in Section 3.1 to state explicitly: BSS is a design-selection criterion for identifying a spatially favorable allocation before data collection. Because the valid BSS allocation space is near-deterministic (2--6 arrangements), permutation-based randomization tests under BSS are infeasible. Investigators using BSS should conduct inference via model-based SAR maximum likelihood estimation, not permutation tests. This is now also stated as an explicit limitation in Section 5.2 (Limitations). The practical value of BSS lies in pre-study design evaluation: before beginning the trial, investigators can compare BSS vs. SRS in terms of expected worst-case estimation error under their anticipated spillover regime and grid geometry---as this simulation study demonstrates.	
C6	According to the results, the bias from these designs can be large. The reviewer argues that 'fried egg' or buffer zone designs should be recommended over BSS, especially under TrtSpill. The results suggest that the proposed designs may not be suitable for practical use.	Future Considerations, new paragraph ~line 1264 in Revisions_V2.tex

Comment (ID)	Reviewer Comment (Summarized)	Location & Evidence
Response:	We acknowledge the reviewer's concern and have added buffer-zone ('fried-egg') designs as an important future direction in Section 5.3 (Future Considerations). We agree that buffer zones represent a complementary and promising strategy: rather than optimising the allocation to minimise cross-boundary spillover at the design stage, buffer zones mitigate contamination at the analysis stage by removing the clusters most likely to be affected. We encourage this line of research and note it explicitly. We respectfully defend the paper's contribution on the following grounds: (1) The primary contribution is characterising the bias landscape across grid geometries and spillover regimes. Knowing that BSS amplifies bias under $\text{TrtSpill} + \rho > 0$ (the checkerboard paradox) is itself a novel and actionable finding that guides design decisions. (2) In TrtNoSpill scenarios---a realistic setting for many interventions where contamination flows only into untreated neighbours---BSS consistently achieves near-zero MSE and variance. (3) Where TrtSpill is anticipated, our results clearly indicate that SRS outperforms BSS (or at minimum practitioners should not use BSS), which is practical guidance. The paper therefore provides actionable conditional recommendations rather than a single universal endorsement of BSS.	

Summary of All Changes

The table below lists all changes made in Revisions_V2.tex for this revision round.

ID	Change Summary	Status
C1	Section 2.3.1, ~line 283 in Revisions_V2.tex	Done
C2	Section 2.4 SAR model, ~line 529; clarification of cluster-level indexing added ~line 527	Done
C3	Section 2.4: after Eq (1) ~line 520; model description ~lines 525--530	Done
C4	Section 3.1 after Equation DataGen, new paragraph ~line 607 in Revisions_V2.tex	Done
C5	Section 3.1 inference paragraph expanded ~line 663; new sentence in Limitations ~line 1224	Done

ID	Change Summary	Status
C6	Future Considerations, new paragraph ~line 1264 in Revisions_V2.tex	Done

Closing

We believe the revised manuscript fully addresses the Reviewer's concerns. We have clarified the estimand definitions, corrected the spillover notation, resolved indexing inconsistencies, explicitly stated the DGP and estimation model, strengthened the inferential framing of BSS, and added buffer-zone designs as a future direction. We hope that these revisions demonstrate the methodological rigor and practical value of our work, and we look forward to hearing from you.

We remain grateful for the Reviewer's expertise and commitment to improving this manuscript.

Sincerely,

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